

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Mining Engineering

2. Subject Name: Safety Engineering and Mine Legislation **L-T-P:** 3-0-0 **Credits:** 3

3. Pre-requisites: Underground Coal Mining, Underground Metal Mining, Surface Mining

4. Syllabus and reference books:

- Mine Safety - a modern approach;
- Accidents and their effects;
- Theories of accident causation: Domino Theory, Human Factors Theory, Multiple causation approach, Epidemiological Theory, Systems Theory, Combination Theory, Behavioral Theory, Safety Analysis Methods and Indices: Hazards and Operability (HAZOP), Job Safety Analysis (JSA), Preliminary Hazard Analysis (PHA), Failure Mode and Effective Analysis, Interface Safety Analysis (ISA), Fault Tree Analysis, Markov Method;
- Risk assessment and management;
- Human Factors and Error in Mine Safety, Drugs and Accident Causation, Depression and Accident Causation, Management Failures and Accident Causation;
- Mine accidents, reports and records;
- Accident investigation and analysis;
- The Occupational Safety, Health and Working Conditions Code, 2020;
- The Occupational Safety, Health and Working Conditions Code(Central) Rules, 2020;
- Central Electricity Authority Regulations, 2010;
- The Coal Mines Regulations, 2017;
- The Metalliferous Mines Regulations, 2019;

Reference Books:

1. Dhillon B.S. (2003). Engineering Safety: Fundamentals, Techniques, and Applications. New York: World Scientific Publishing.
2. Dhillon B.S. (2010). Mine Safety: A Modern Approach, Springer 186 pp.
3. David L. Goetsch (2015). Occupational Safety and Health for Technologists, Engineers, and Managers, Pearson Education Limited (UK).
4. Occupational Safety, Health And Working Conditions Code, (2020). Professional Book Publishers, New Delhi
5. The Occupational Safety, Health and Working Conditions Code (Central) Rules, (2020). Ministry of labour and employment, Government of India

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1	Mine Safety - a modern approach;	2
	Accidents and their effects;	2
	Theories of accident causation: Domino Theory, Human Factors Theory, Multiple causation approach, Epidemiological Theory, Systems Theory, Combination Theory, Behavioral Theory, Safety Analysis Methods and Indices: Hazards and Operability (HAZOP), Job Safety Analysis (JSA), Preliminary Hazard Analysis (PHA), Failure Mode and Effective Analysis, Interface Safety Analysis (ISA), Fault Tree Analysis, Markov Method; Risk assessment and management;	10
	Human Factors and Error in Mine Safety, Drugs and Accident Causation, Depression and Accident Causation, Management Failures and Accident Causation;	2
	Mine accidents, reports and records;	2
	Accident investigation and analysis;	2
	The Occupational Safety, Health and Working Conditions Code, 2020;	4
	The Occupational Safety, Health and Working Conditions Code(Central) Rules, 2020;	2
	Central Electricity Authority Regulations, 2010;	2
	The Coal Mines Regulations, 2017;	3
	The Metalliferous Mines Regulations, 2019;	3
Total number of hours		34